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Broadening the Domain of Application of Regression Models:
Nonlinear Relationships

This file is named: 06_slides.pptx
Commands are in: 06_notebook.ipynb

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Derivation of Elasticity in log-log model
Derivation for semi-log model with small change in an explanatory variable




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Regression Models with Nonlinear Relationships

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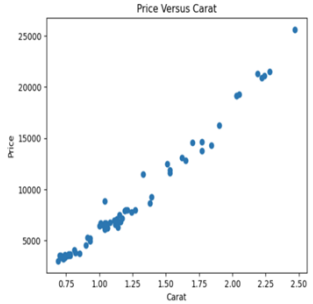
## *Functional Forms*

- Not everything is linear. But often nonlinear relationships can be recast in a form that permits use of regression analysis.




We have 67 diamonds with much more size variation than in our earlier analysis of 17 diamonds. These data are in file Diamonds_67. All have the same color, cut, and clarity

Looks like there is a curve!

Data Transformation can help us better fit the data.

The way in which we transform the data is called the **functional form.**



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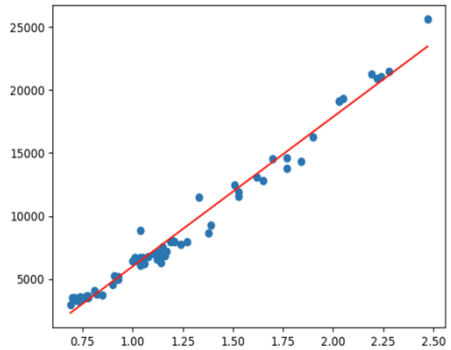
## Fit of Linear Model to The Data

This graph shows the fit of the data to a linear model.

We see that the observations for the smallest and the largest diamonds tend to be above regression the line.

The diamonds of intermediate size tend to be below the line.

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## Quadratic Transformation: Diamonds 67

We can approximate the nonlinear relationship between y and x for diamonds using a quadratic: $y=\beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$

```
# Estimate Price vs Carat and Carat^2.
# Note the expression for the square of Carat
QuadDiam = tprstats.model("cs", "Price~Carat+ {Carat**2}", Diamonds_67)
QuadDiam.summary()
```

This is how you put the square of a variable in a regression.

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | -1536.9965 | 577.584 | -2.661 | 0.010 | -2690.852 | -383.141 |
| **Carat** | 5043.4539 | 960.427 | 5.251 | 0.000 | 3124.780 | 6962.128 |
| **Carat ** 2** | 2348.4362 | 328.410 | 7.151 | 0.000 | 1692.362 | 3004.511 |

Note from the p-value that the coefficient of the squared term is very significant.
This provides strong evidence that the relationship is in fact nonlinear.
We will study a statistical test called the Ramsey test that generalizes this idea.

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## Plot of Quadratic Fit to The Data

The quadratic approximation appears to provide a good fit to the data.

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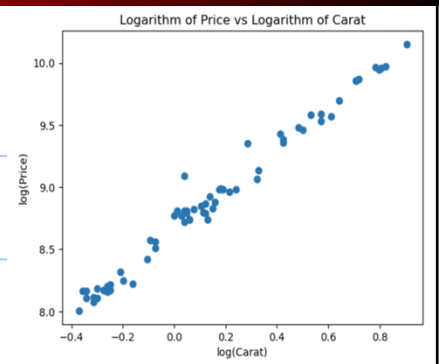
## Logarithmic Transformation

With both price and carat expressed as natural logarithms, the relationship appears to be linear.

```
# Plot log of diamond Price against log of Carat
plt.scatter(numpy.log(Carat), numpy.log(Price))
plt.title("Logarithm of Price vs Logarithm of Carat")
plt.xlabel("log(Carat)")
plt.ylabel("log(Price)")
```

Note when using the plt command that we use numpy.log() to calculate the natural logarithm of a variable.

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## Regression with variables in log form

```
# Estimate log-log model for Diamonds_65
# Interpret coefficient of log(Carat)
RegLogDiam67 = tprstats.model("cs", "log(Price)~log(Carat)", Diamonds_67)
RegLogDiam67.summary()
```

Note we use log() without numpy in the regression command.

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | log(Price) | **R-squared:** | 0.982 |
| **Model:** | OLS | **Adj. R-squared:** | 0.982 |
| **No. Observations:** | 67 | **F-statistic:** | 7871. |
| **Covariance Type:** | HC1 | **Prob (F-statistic):** | 1.51e-69 |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | 8.6596 | 0.011 | 795.957 | 0.000 | 8.638 | 8.681 |
| **log(Carat)** | 1.6514 | 0.019 | 88.720 | 0.000 | 1.614 | 1.689 |

Interpretation:

- When both the independent and dependent variables are in log form, the coefficient of the independent variable is an elasticity. This is a very nice feature of a log-log relationship.
- In this example, the estimated elasticity of Price with respect to Carat is 1.65: the predicted price increases by 1.65% for each 1% increase in carat.
- This link takes you to the derivation in the optional appendix. Then, to return to this slide click Back.

*You generally do not want to use the "Scaled Coefficients" feature to calculate elasticity when your variables are transformed. The "Scaled Coefficients" elasticity should only be used when variables are not transformed.*

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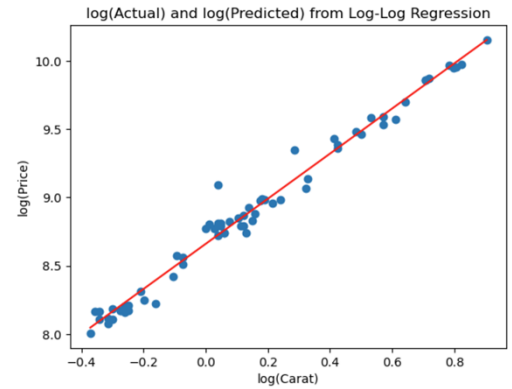
## Fit of Log-Log Model to Data

We see that the log-log model fits the data well.

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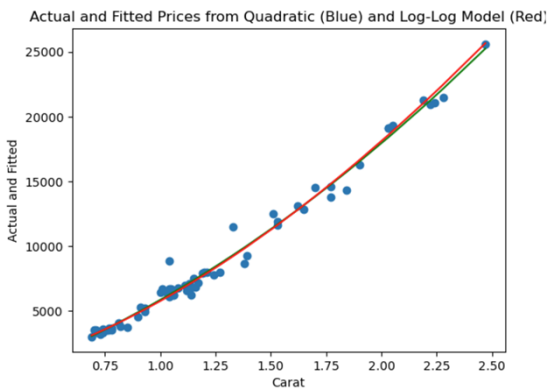
## Comparison of Quadratic and Log-Log

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## Which to use?

The two transformations give very similar fit within the range of the data.

We would tend to prefer the log-log model because:

- We need one fewer coefficients to fit the data.
- The constant elasticity property of the log-log model has much intuitive appeal: the elasticity is the same everywhere along the curve for a log-log function.

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Prediction for Models with Dependent Variable in Log Form

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## *Predicting Values for A Dependent Variable in Log Form*

A customer is considering purchase of a diamond with Carat weight 2 and would like to predict the price.
The Clarity, Color, and Cut are the same as with the diamonds in our data set (Clarity=4, Color=11, Cut=2).
When the dependent variable is in log form, we have two options for prediction.

1. We can predict the log of the dependent variable, log(Price)

```
# Predicted log(Price) and prediction interval for Carat=2 using log-log regression
Xnew = pandas.DataFrame({"Carat": [2]})
RegLogDiam67.prediction_intervals(Xnew)
```

| Predicted | Lower | Upper |
|---|---|---|
| 9.804282 | 9.641797 | 9.966766 |

2. In most applications, we would want to predict the value of the dependent variable rather than the log of the value. To do so, we use the following

```
# Predicted Price and prediction interval for Carat=2 using log-log regression
RegLogDiam67.prediction_intervals(Xnew).map(numpy.exp)
```

We predict that the diamond will sell for $18,111 with 95% prediction interval [$15395, $21306].

| Predicted | Lower | Upper |
|---|---|---|
| 18111.125492 | 15394.985664 | 21306.474311 |

The same logic applies to calculating fitted values.
We calculate the fitted values of log price:

```
# Calculate fitted log prices.
Log_yhat = RegLogDiam67.predict()
```

We then exponentiate to obtain the fitted values of price:

```
# Calculate fitted Prices from log model
yhat = numpy.exp(Log_yhat)
```

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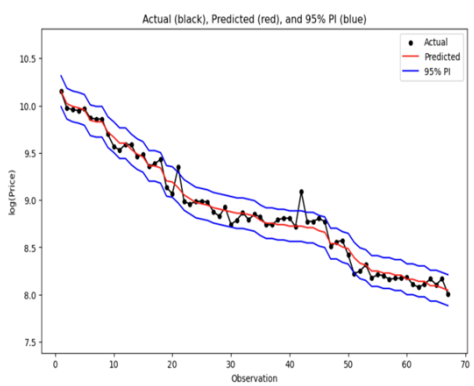
## *View Actual, Predicted, and Prediction Intervals*

A useful diagnostic is provided by the following command:

```
# Plot actual and fitted log prices
 and 95% prediction intervals.
RegLogDiam67.plot_actual_fitted()
```

The above command produces the plot to the right.

- We see that one observation is well above the upper bound of its 95% prediction interval.
- We would investigate whether there is a coding error for such an observation.
- In this case there is not a coding error.
- The jeweler negotiated a high price for this diamond.

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## *What To Learn From This Application*

This application shows you how to:

- Do a scatter plot with logarithms of variables
- Estimate a model with variables in log form
- Do a plot show actual and fitted log values and 95% prediction intervals
- Do a prediction and form a prediction interval with variables in log form
- Exponentiate the preceding to obtain the prediction and prediction interval in natural (unlogged) units.

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Models with A Quadratic Functional Form for an Explanatory Variable

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## *Application: Auto Fuel Efficiency and Speed*

Above we studied an application where a log-log model works well.

We now study and application where a quadratic model works well.

An automobile manufacturer is studying the relationship between fuel efficiency and speed for its best-selling automobile. The manufacturer has conducted the following data collection exercise:

- The car is driven at a uniform speed of 20 miles per hour (mph) on a test track, and the fuel efficiency in miles per gallon (mpg) is recorded.
- The same route is driven at 21 mph, then 22 mph, …, up to 69 mph, with fuel usage (mpg) recorded at each speed.

Data file Fuel_Efficiency_vs_Speed.xlsx contains the resulting data on speed and miles per gallon.

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## *Linear Regression of MPG versus Speed*

The commands below perform calculations that generate the output on the next slide. The following two commands estimate a linear regression of MPG versus speed and display the regression output.

```
# Estimate a linear model of MPG vs Speed
LinMPG = tprstats.model("cs", "MPG~Speed", MPGData)
LinMPG.summary()
```

The following calculates fitted values from the regression and names the result LinFit.

```
# Calculate fitted values from linear regression and name LinFit
LinFit = LinMPG.predict()
```

The following commands extract MPG and Speed from the data file.

```
# For convenience, extract these variables.
MPG = MPGData["MPG"]
Speed = MPGData["Speed"]
# We can now  use the variables without referencing the data file
```

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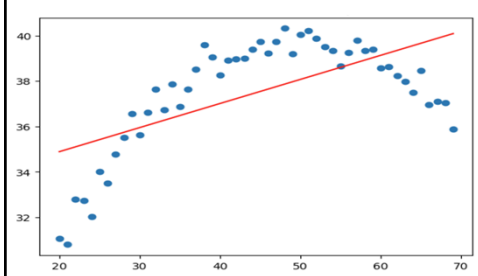
## *Linear Regression of MPG versus Speed*




Coefficient of Speed is very significant, but the fit is obviously poor.

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | 32.7673 | 1.128 | 29.055 | 0.000 | 30.500 | 35.035 |
| **Speed** | 0.1061 | 0.024 | 4.409 | 0.000 | 0.058 | 0.154 |

- The relationship is clearly not linear.
- A log form will not work because a log function is monotone: i.e., log(x) increases as x increases.
- So a log function cannot fit a ∩ or ∪ shape.
- Quadratic is a promising possibility.

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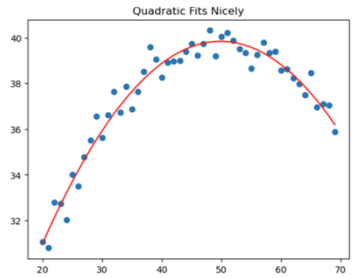
## Quadratic Function Fit To Fuel Efficiency Data

```
# Estimate a quadratic in MPG vs Speed
# You can use either I(Speed**2) or {Speed**2}
QuadMPG = tprstats.model("cs", "MPG ~ Speed + {Speed**2}", MPGData)
QuadMPG.summary()
```

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | MPG | **R-squared:** | 0.958 |
| **Model:** | OLS | **Adj. R-squared:** | 0.956 |
| **No. Observations:** | 50 | **F-statistic:** | 451.7 |
| **Covariance Type:** | HC1 | **Prob (F-statistic):** | 2.06e-31 |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | 15.1718 | 0.779 | 19.479 | 0.000 | 13.605 | 16.739 |
| **Speed** | 0.9899 | 0.035 | 27.905 | 0.000 | 0.918 | 1.061 |
| **Speed ** 2** | -0.0099 | 0.000 | -25.966 | 0.000 | -0.011 | -0.009 |

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## Interpreting Quadratic Model

The effect on fuel efficiency of increasing speed by one mile per hour depends on current speed. From the plot on the previous slide, we see that:

- An increase in speed from 20 mph to 21 mph improves fuel efficiency.
- An increase in speed from 65 mph to 66 mpg reduces fuel efficiency.
- The slope of the function depends on the speed of the car.

For the quadratic model, $\hat{y} = b_0 + b_1 x + b_2 x^2$

- The derivative tells us the slope is: $\frac{d\hat{y}}{dx} = b_1 + 2b_2 x = .99 - 2 * .0099 * x$

Thus, the slope for our fuel efficiency regression is .99-2*.0099*speed

Using this equation for the derivative, we calculate the following:

- If current speed is 20, a one mph increase in speed will increase fuel efficiency by (.99-2*.0099*20)=.59 mpg.
- If current speed is 65, a one mph increase in speed will reduce fuel efficiency by .30 mpg, (.99-2*.0099*65)= -.30 mpg.

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## What is the most fuel-efficient speed?

From the previous slide, the derivative is: $\frac{d\hat{y}}{dx} = b_1 + 2b_2 x$

To find the maximum, we set the derivative equal to zero: $b_1 + 2b_2 x = 0$ . Then solve for x.

We find that the most fuel-efficient speed is 50 miles per hour.

$$x^* = -\frac{b_1}{2b_2} = \frac{.99}{2 \cdot .0099} = 50$$

Calculate predicted fuel efficiency at the above speed. The result, shown below, is 39.8 mpg.

$$\hat{y} = b_0 + b_1 x + b_2 x^2 = 15.17 + .9898 * 50 - .00993 * 50 \wedge 2 = 39.8$$

Of course, most drivers care about how long their trip takes as well as fuel economy. We investigate further on the next slide.

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## Prediction and Prediction Interval

Suppose the car is driven at 70 miles per hour. Calculate the predicted miles per gallon:

$$\hat{y} = b_0 + b_1 x + b_2 x^2 = 15.17 + .9898 * 70 - .00993 * 70 \wedge 2 = 35.8$$

Predicted fuel economy drops from 39.8 to 35.8 with speed increased from 50 to 70 MPH.

Let's use Python to do the above prediction and find a 95% prediction interval.

```
# Predict mpg if speed=70 and form prediction interval. Note in this command
# that you need only enter the value for Speed. In doing the prediction,
# the software automatically calculates and includes the term in Speed^2.
Xnew = pandas.DataFrame({"Speed": [70]})
QuadMPG.prediction_intervals(Xnew)
```

We find the predicted fuel efficiency is 35.8, as we calculated above.

We find there is a 95% probability that fuel economy will be in the interval [34.7, 36.9]

| | Predicted | Lower | Upper |
|---|---|---|---|
| **0** | 35.805761 | 34.689577 | 36.921945 |

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## *Summary Regarding Quadratic Model*

Our investigation of fuel economy as a function of speed nicely illustrates application of a quadratic model.

We have seen how to:

- Estimate a quadratic model
- Find the slope at any point on the function
- Find the value that maximizes the function
- Do a prediction and calculate a prediction interval

In our fuel economy application, the estimated function has a $\cap$ shape.

In other applications you may obtain a $\cup$ shape. In such a case, you will obtain a minimum when you set the derivative equal to zero.



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Semi-Log Models



## *Definition of a Semi-log Relationship*

Definition: When the dependent variable is in log form and the independent variable is not logged, we have a semi-log relationship.

Using our standard notation for the population regression model, we write the semi-log model as: $\text{Log}(y) = \beta_0 + \beta_1 x$

Of course, we will add an error term when we turn to estimation.

Example Suppose you invest an amount $y_0$ at time zero with continuous compounding at interest rate r.

At time t, it's value is: $y = y_0 e^{rt}$

If we take logarithms of both sides of the above exponential function, we obtain:

$\text{Log}(y) = \log(y_0) + rt$

In this investment example, we know that the coefficient r is (approximately) the fraction by which y increases in value when t increases by one unit.



## *Interpreting Coefficient in a Semi-log Relationship*

Our equation is:

1. $\text{Log}(y) = \beta_0 + \beta_1 x$

Suppose x increases by $\Delta x$. The increase in $\Delta y$ must satisfy:

2. $\text{Log}(y+\Delta y) = \beta_0 + \beta_1(x+\Delta x)$

Subtract Equation (1) from Equation (2):

3. $\text{Log}(y+\Delta y) - \text{Log}(y) = \beta_1 \Delta x$

The difference in logs of variables can be written as the log of the ratio of the variables. Hence, we can rewrite (3) as:

4. $\log\left(\frac{y+\Delta y}{y}\right) = \beta_1 \Delta x$

Exponentiate both sides of (4) and simplify as: 5. $\left(1+\frac{\Delta y}{y}\right) = e^{\beta_1 \Delta x}$

Subtract 1 from both sides of 5 and rewrite as: 6. $\frac{\Delta y}{y} = (e^{\beta_1 \Delta x} - 1)$

The formula in Equation (6) gives us the exact fraction by which y changes when x changes by amount $\Delta x$.

## Interpreting Semi-Log Coefficient Continued

Suppose $\beta_1$ =.35. Suppose x changes by one unit: $\Delta x$=1.
The formula in Equation (6) from the previous slide is:

$$\frac{\Delta y}{y} = (e^{\beta_1 \Delta x} - 1)$$

Substituting $\beta_1$ = .35 and $\Delta x$=1 into the above, we obtain:

$$\frac{\Delta y}{y} = (e^{.35*1} - 1) = .419$$

- Thus, y increases by fraction .419 when $\beta_1$ =.35 and x increases by one unit.
- Equivalently we can say y increases by 41.9% when x increases by one unit.

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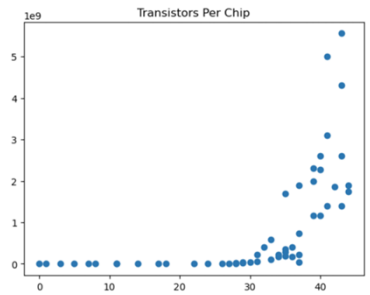
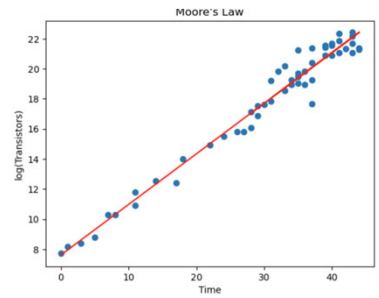
## Application Intel and Moore's Law

The graphs below show transistors per Intel chip from 1970 through 2015. The left is unlogged. The right is the semi-log with the least squares regression equation shown.

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## Regression and Interpretation

By what percentage did transistors per chip increase per year:

$$\frac{\Delta y}{y} = (e^{.3369*1} - 1) = .399$$

Answer: 40%

By what percentage did transistors per chip increase per two year interval:

$$\frac{\Delta y}{y} = (e^{.3369*2} - 1) = .958$$

Answer: 96%

```
# Estimate Moore's law regression
MooreReg = tprstats.model("cs", "log(Transistors)~Time", Moores_law_intel)
MooreReg.summary()
```

| | | | |
|---|---|---|---|
| Dep. Variable: | log(Transistors) | R-squared: | 0.969 |
| Model: | OLS | Adj. R-squared: | 0.968 |
| No. Observations: | 53 | F-statistic: | 3223. |
| Covariance Type: | HC1 | Prob (F-statistic): | 9.06e-48 |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| Intercept | 7.6034 | 0.149 | 50.940 | 0.000 | 7.304 | 7.903 |
| Time | 0.3369 | 0.006 | 56.774 | 0.000 | 0.325 | 0.349 |

Note: In 1965 Moore predicted a doubling (i.e., 100% increase) each two years. Our regression result shows that his prediction was astonishingly close.

He had a very remarkable career: https://en.wikipedia.org/wiki/Gordon_Moore

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## Interpreting Semi-Log Coefficient Continued

We next provide a convenient approximation that can be used when $\beta_1\Delta x$ is fairly small (.15 or less).

If $\beta_1\Delta x$ is fairly small (.15 or less), y changes by fraction $\beta_1\Delta x$ when x changes by amount $\Delta x$. The derivation is provided in an optional appendix to these slides.

- For example, suppose $\beta_1$ = .05. Suppose x increases by one unit, i.e., $\Delta x$=1. Then $\beta_1\Delta x$ = .05. This is less than .15.
- If $\beta_1$ = .05, a one unit increase in x increases y by fraction .05 or, equivalently, by 5%.
- Consider the investment equation on the second slide of these notes. If the interest rate r=.05, then the semi-log coefficient is .05, the value of the investment increases by 5% per year.
- What is the approximate growth of transistors per chip over a three-month period? $\beta_1\Delta x$ =.3369*(1/4) = .084. Hence approximately 8.4%.
- Compare to exact calculation: $\frac{\Delta y}{y} = (e^{.3369*(1/4)} - 1) = .088$, **or 8.8%**

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## *Summary of Interpretation with Both Log-Log and Semi-Log*

The following model has both semi-log and log-log forms: $\log(y)=\beta_0 + \beta_1 x_1 + \beta_2 \log(x_2)$

Interpreting $\beta_1$. If $\beta_1 < .15$, you can use the following approximation:

- $\beta_1$ is the **fraction** by which y changes when $x_1$ increases by **one unit**, or, equivalently,
- $100*\beta_1$ is the **percent** by which y increases when $x_1$ increases by **one unit** (sometimes called a semi-elasticity)

If $\beta_1 > .15$, use the exact calculation:

- $(e^{\beta_1}-1)$ is the **fraction** by which y changes when $x_1$ increases by **one unit**, or
- $100*(e^{\beta_1}-1)$ is the **percent** by which y increases when $x_1$ increases by **one unit** (sometimes called a semi-elasticity)

Interpreting $\beta_2$:

- $\beta_2$ is the **percent** by which y increases when $x_2$ increases by **one percent** (i.e., elasticity).

**Please memorize this slide.**

**You will be very glad you did!**

**And I will too!**

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## *Summary for Semi-log Relationship*

- We defined a semi-log relationship.
- We worked through interpretation of the coefficient in the semi-log relationship.
- We did an application using data for transistors per Intel chip and the relationship to Moore's Law.

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Nonlinear Relationships: Ramsey Test

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## *Detecting and Evaluating Nonlinear Relationships*

Thus far, we have seen three types of transformations and we have applied each of them:

- Log-log
- Semi-log
- Quadratic

In these slides:

- We develop a statistical test, the Ramsey test, to help us determine whether we need a transformation of our data.
- We study a visual diagnostic that can sometimes help us to determine which transformations we might consider.

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## Testing Functional Form

I created a data set named: *Linear_Model_Example_Data.xlsx*

- The dependent variable is named y.
- There are two right-hand-side variables named x1 and x2.

I imported the data and renamed the file *MyDat*. I then estimated the following regression:

```
RegExample = tprstats.model("cs", "y~x1+x2", MyDat)
RegExample.summary()
```

I calculated the fitted values and residuals:

```
yhat=RegExample.fittedvalues
resid=RegExample.resid
```

I then plotted resid vs yhat with a horizontal line at zero.

The resulting plot is on the next slide.

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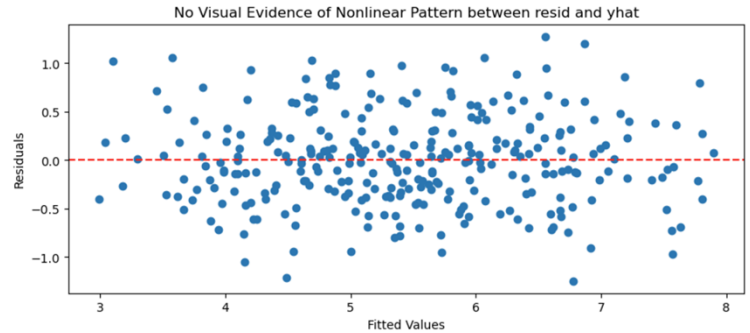
## Plot of resid vs yhat

We see that there is relatively uniform scatter across the range of yhat.

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## Further Analysis of The Example on the Previous Page

If our regression of y versus x1 and x2 captures the relationship in the data, there should not be a nonlinear relationship between resid and yhat.

One way to investigate this is to estimate a quadratic regression of resid versus yhat and yhat^2.

I created a data frame with the residuals, yhat, and the square of yhat as follows:

```
MyDat2 = pandas.DataFrame({ 'resid': resid, 'yhat': yhat, 'yhatsq': yhat**2 })
```

This data frame is used on the next slide.

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## Further Analysis of The Example on the Previous Page

I estimated a regression with resid versus yhat and yhatsq

```
ResidReg = tprstats.model("cs", "resid~yhat+yhatsq",data=MyDat2)
ResidReg.summary()
```

We find insignificant coefficients for yhat and yhatsq.

We also find and a p-value=.987 for the overall regression indicating that the variables are not jointly significant. This is p-value for the null hypothesis that all coefficients are zero except the intercept.

Thus, we find no evidence of a nonlinear pattern between resid and yhat for my made-up example.

We conclude that the linear model is a good choice for my made up example.

| | | | |
|---|---|---|---|
| Dep. Variable: | resid | R-squared: | 0.000 |
| Model: | OLS | Adj. R-squared: | -0.007 |
| No. Observations: | 299 | F-statistic: | 0.01306 |
| Covariance Type: | HC1 | Prob (F-statistic): | 0.987 |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| Intercept | -0.0926 | 0.592 | -0.157 | 0.876 | -1.257 | 1.072 |
| yhat | 0.0352 | 0.219 | 0.160 | 0.873 | -0.396 | 0.466 |
| yhatsq | -0.0032 | 0.020 | -0.162 | 0.872 | -0.042 | 0.036 |

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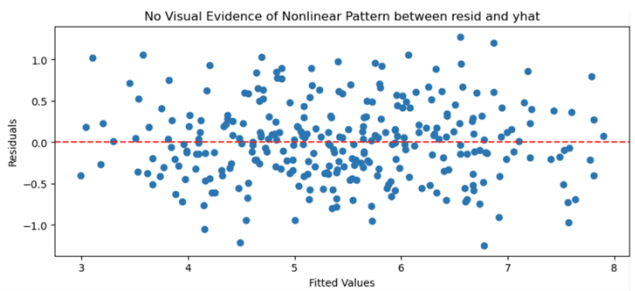
## Plot for Example on Previous Slides

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## Analyzing Functional Form For Diamond Data

Let's do the same analysis *Diamonds_67.xlsx*

We estimate a linear model of Price versus Carat.

```
# Linear model of Diamonds_67
LinDiam = tprstats.model("cs", "Price~Carat", Diamonds_67)
LinDiam.summary()
```

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | Price | **R-squared:** | 0.979 |
| **Model:** | OLS | **Adj. R-squared:** | 0.978 |
| **No. Observations:** | 67 | **F-statistic:** | 2074. |
| **Covariance Type:** | HC1 | **Prob (F-statistic):** | 4.88e-51 |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | -5889.1329 | 299.710 | -19.649 | 0.000 | -6487.695 | -5290.570 |
| **Carat** | 1.188e+04 | 260.897 | 45.541 | 0.000 | 1.14e+04 | 1.24e+04 |

At first look, this seems like a very good model.
The coefficient of Carat is highly significant.
The R-squared is .98.
But looks can be deceiving, as we next show.

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## Analyzing Functional Form For Diamond Data

Let's do the same analysis Diamonds_67.xlsx as we did with my made up data.
We estimated the linear model of Price versus Carat in previous slide.
We then extract the fitted values and residuals and name them yhat and resid.
We fit a quadratic function of residuals versus yhat and the square of yhat.
We then create plot on next slide. All of this is done in the commands below.

```
# View plot of resid vs yhat with quadratic fit
resid=LinDiam.resid
yhat=LinDiam.predict()
plt.figure(figsize=(10, 5))
plt.scatter(yhat, resid)
df = pandas.DataFrame({ 'resid': resid, 'yhat': yhat, 'yhatsq': yhat**2 })
residreg=tprstats.model("cs","resid~yhat + yhatsq", df)
reshat=residreg.predict()
plt.plot(yhat, reshat, color="red")
plt.axhline(y=0, color='lightblue', linestyle='--')
plt.title("Estimated Residuals Versus Fitted Values from Linear Regression with 67 Diamonds")
plt.xlabel("yhat")
plt.ylabel("resid")
plt.show()
```

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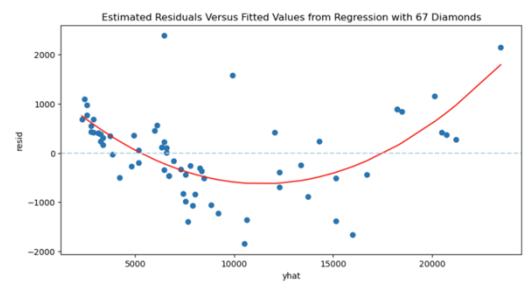
## Plot of resid vs yhat for Linear Model of Diamonds 67

We observed a pronounced nonlinear ∪-shaped pattern in the plot.
There are many positive residuals to the left and to the right and mostly negative residuals in the middle.




Note: You do not need to learn how to do this plot. I did the plot to help you understand the Ramsey test that we will now study.

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## Automating The Test of Functional Form

A test of functional form very similar to that illustrated on the previous slides is implemented in Python by the Ramsey test as follows:

```
# Ramsey reset test
LinDiam.ramsey_test()
```

We obtain:

| power | pvalue |
|---|---|
| 2 | 1.828668e-08 |
| 3 | 1.428546e-07 |

The Ramsey test reports the overall regression p-value for two regressions. One employs a quadratic function as illustrated on the previous slides. The other employs a cubic function. A low p-value for either is evidence against the chosen functional form.

Note: power here is not the concept of power we used in experimental design. Rather it denotes yhat raised to the second power and the third power.

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## Further Application of The Ramsey Test

The Ramsey test is a test of the functional form we have estimated (e.g., our linear regression of Price versus Carat.)

If we obtain low p-values, we have evidence that our chosen functional form is not fitting the data well.

When we began our discussion of functional form, we considered the data in Diamonds_67.

In that earlier analysis, we saw from a plot of the data and regression line that a linear model did not fit well.

We then tried a log-log model. When we plotted the estimated model with the data, it appeared to fit well.

On the previous slide, we applied the Ramsey test to the model linear in Price and Carat.

Let's now do the test with the log model.

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## Regression with variables in log form

```
# Estimate log-log model for Diamonds_67
LogDiam67 = tprstats.model("cs", "log(Price)~log(Carat)", Diamonds_67)
LogDiam67.summary()
```

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | 8.6596 | 0.011 | 795.957 | 0.000 | 8.638 | 8.681 |
| **log(Carat)** | 1.6514 | 0.019 | 88.720 | 0.000 | 1.614 | 1.689 |

```
# Ramsey test of log model of Diamonds_67
LogDiam67.ramsey_test()
```

We obtain high p-values for both terms. Hence, the Ramsey test does not reject our functional form. This is favorable evidence for our functional form.

| | power | pvalue |
|---|---|---|
| **0** | 2 | 0.220270 |
| **1** | 3 | 0.409141 |

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## Recap: Ramsey Test with Diamonds_67 Data

The results of our Ramsey tests from the previous slides are summarized here.

The very low p-values here suggest linear model fits poorly.

| power | pvalue |
|---|---|
| 2 | 1.828668e-08 |
| 3 | 1.428546e-07 |

The relatively high p-values here suggest the log-log model fits well.

| | power | pvalue |
|---|---|---|
| **0** | 2 | 0.220270 |
| **1** | 3 | 0.409141 |

Conclude that the linear function does not fit the data well.

The log-log function appears to provide a good fit to the Diamonds_67 data.

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## *Another Application of The Ramsey Test*

Let's investigate the function we used for our Tire Pricing model using the

```
# Ramsey test of tire model
TireReg.ramsey_test()
```

| power | pvalue |
|---|---|
| 2 | 0.130708 |
| 3 | 0.300243 |

We obtain p-values of .13 and .30. Hence, this functional form appears to be satisfactory, although the p-value with one fitted term is a bit low.

We will investigate this a bit more in the future.





## *Practice Using The Ramsey Test*

As further practice, you might want to investigate using the data set Fuel_Efficiency_vs_Speed

Import the data set and try the following:

- Estimate the linear model of MPG vs Speed and then apply the Ramsey test.
- Estimate the quadratic model and apply the Ramsey test to that model.





## *Comments on Ramsey Test*

- The Ramsey test is a decision aid.
- It tests the function you have chosen. It does not tell you what function to use.
- When you have a large number of observations, it may be difficult to find a functional form that is not rejected by the Ramsey test.
- You may decide to use a functional form even if it is rejected by the Ramsey test, however…
- Rejection by the Ramsey test suggests that you should do some investigation of alternative functional forms.
- You do not want to be too slavish in trying to satisfy the Ramsey test.
- With time series data, the Ramsey test can often be uninformative, particularly if the error terms are correlated with each other.
- With all of the above caveats in mind, the Ramsey test often provides helpful information in choosing functional form.





*Statistical Decision Making*
*45-752*

Nonlinear Relationships: Leverage Plots

## *Leverage Plots (also called Added-Variable Plots)*

Leverage plots are designed to present information in scatter plot form for models with multiple right-hand-side variables.

For simple regression, a leverage plot provides the same information as a scatter plot of y versus x.

Intuitively, the leverage plot is designed to provide similar information when there are several right-hand-side variables.

The leverage plot shows the effect of each right-hand-side variable on y after removing the effects of all the other X variables.

The appendix has two slides with more information. Click Back on the second appendix slide to return here. More Leverage Plot Detail



## *Leverage Plot Application*

Let's study the Diamonds_211 data set, which has diamonds that vary in Carat and Clarity

- Consider a linear regression of Price versus Carat and Clarity
- Estimate a linear model and verify that the Ramsey test strongly rejects this linear model.

Create Leverage Plots (also called added-variable plots).

The following command does a plot of Price versus Carat holding Clarity constant.

```
sm.graphics.plot_partregress('Price', 'Carat', ['Clarity'], data=Diamonds_211, obs_labels=False)
plt.show()
```

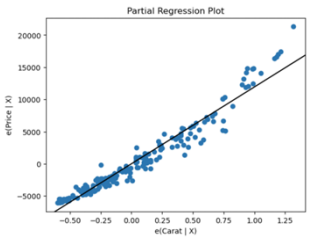
The following command does a plot of Price versus Clarity holding Carat constant.

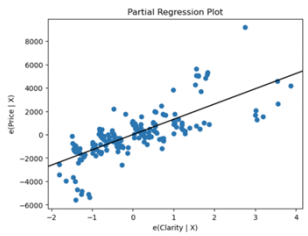
```
sm.graphics.plot_partregress('Price', 'Clarity', ['Carat'], data=Diamonds_211, obs_labels=False)
plt.show()
```

The two plots are on the next slide.



## *Leverage Plots*




From the left plot, we see clear evidence of a nonlinear relationship of price and carat.

- This is very typical of a log-log pattern.
- Let's try taking the logarithms of price and carat. We will then view the leverage plots from that model.



## *Model for 211 Diamonds Varying in Carat and Clarity*

```
# Estimate model with log(Price) and log(Carat)
LogDiam211 = tprstats.model("cs", "log(Price) ~ log(Carat) + Clarity", Diamonds_211)
LogDiam211.summary()
```

OLS Regression Results

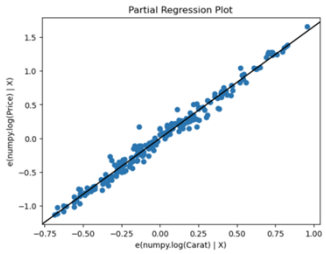
| | | | |
|---|---|---|---|
| **Dep. Variable:** | log(Price) | **R-squared:** | 0.984 |
| **Model:** | OLS | **Adj. R-squared:** | 0.983 |
| **No. Observations:** | 211 | **F-statistic:** | 8249. |
| **Covariance Type:** | HC1 | **Prob (F-statistic):** | 7.90e-199 |

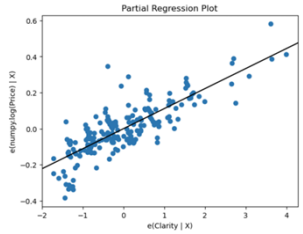
| | **coef** | **std err** | **t** | **P>\|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | 8.1878 | 0.026 | 315.824 | 0.000 | 8.137 | 8.239 |
| **log(Carat)** | 1.6570 | 0.013 | 128.323 | 0.000 | 1.632 | 1.683 |
| **Clarity** | 0.1111 | 0.006 | 19.511 | 0.000 | 0.100 | 0.122 |

We next study leverage plots for this model.

We then turn to interpretation of the coefficients.

## Model with Logs of Price and Carat and Linear in Clarity




We see from the left plot that this model has nicely resolved the nonlinearity with respect to Carat.

For Clarity, the picture is less "clear". We'll first interpret the coefficients in the model we estimated. Then we'll investigate Clarity in more detail.

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## Interpreting the Coefficients

Here we reproduce the regression from previous slides.

```
# Estimate model with log(Price) and log(Carat)
LogDiam211 = tprstats.model("cs", "log(Price) ~ log(Carat) + Clarity", Diamonds_211)
LogDiam211.summary()
```

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| Intercept | 8.1878 | 0.026 | 315.824 | 0.000 | 8.137 | 8.239 |
| log(Carat) | 1.6570 | 0.013 | 128.323 | 0.000 | 1.632 | 1.683 |
| Clarity | 0.1111 | 0.006 | 19.511 | 0.000 | 0.100 | 0.122 |

The coefficient of log(Carat) tells us the elasticity of price with respect to carat is 1.66.

- A one percent increase in the carat weight of a diamond, holding clarity constant, increases predicted price by 1.66%.
- The coefficient is the elasticity (i.e., we do **not** multiply the coefficient by 100 to convert to percent).

The Coefficient of Clarity tells us that, holding carat constant, a one unit increase in clarity increases predicted price by fraction of approximately .111, or, equivalently, by 11.1%.

The Ramsey test for this model yields the following.

The low p-value with one fitted term suggests doing more investigation.

| power | pvalue |
|---|---|
| 2 | 0.108773 |
| 3 | 0.223342 |

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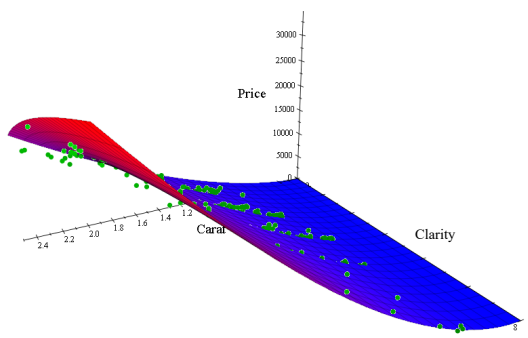
## Plot of Our Estimated Model

Below is a plot of the model on previous slide with data shown in green dots:

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## Model with Separate Indicator Variables for Clarity

Estimate and view model:

```
# Estimate with categories for clarity
LogFacDiam211 = tprstats.model("cs", "log(Price) ~ log(Carat) + C(Clarity)", Diamonds_211)
LogFacDiam211.summary()
```

| | | | |
|---|---|---|---|
| Dep. Variable: | log(Price) | R-squared: | 0.987 |
| Model: | OLS | Adj. R-squared: | 0.987 |
| No. Observations: | 211 | F-statistic: | 3664. |
| Covariance Type: | HC1 | Prob (F-statistic): | 1.00e-204 |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| Intercept | 8.2714 | 0.024 | 338.584 | 0.000 | 8.223 | 8.320 |
| log(Carat) | 1.6714 | 0.012 | 137.116 | 0.000 | 1.647 | 1.695 |
| C(Clarity)[T.3] | 0.2641 | 0.025 | 10.610 | 0.000 | 0.215 | 0.313 |
| C(Clarity)[T.4] | 0.3853 | 0.025 | 15.145 | 0.000 | 0.335 | 0.435 |
| C(Clarity)[T.5] | 0.4634 | 0.026 | 17.850 | 0.000 | 0.412 | 0.515 |
| C(Clarity)[T.6] | 0.5368 | 0.029 | 18.310 | 0.000 | 0.479 | 0.595 |
| C(Clarity)[T.8] | 0.8032 | 0.046 | 17.397 | 0.000 | 0.712 | 0.894 |

- Test functional form:

```
# Ramsey test now gives high p-values for both 1 and 2 fitted terms
LogFacDiam211.ramsey_test()
```

| power | pvalue |
|---|---|
| 2 | 0.628253 |
| 3 | 0.867617 |

- High p-values suggest this is a good functional form.

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## Interpret Clarity Values in The Regression on Previous Slide

The reference category is Clarity=2.

Let's interpret the coefficient of Clarity =3:

Holding constant Carat, a Clarity 3 diamond is predicted to sell for 100*(exp(.264)-1) = 30.2% more than a Clarity 2 diamond.

Interpret the coefficient of Clarity =6:

Holding constant Carat, a Clarity 6 diamond is predicted to sell for 100*(exp(.5368)-1) = 71% more than a Clarity 2 diamond.

For how much more is a Clarity 8 diamond predicted to sell than a Clarity 5 diamond of the same Carat?

100*(exp(.803-.463)-1) = 40.49%

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## Testing Hypothesis about Coefficients of Factors

Suppose we wish to test the null hypothesis that the population coefficients of Clarity 3 and Clarity 4 are equal.

```
# Test null hypothesis that population coefficient of Clarity=3 is same as Clarity=4
LogFacDiam211.wald_test("C(Clarity)[T.3] = C(Clarity)[T.4]")
```

The p-value is 0 to many significant figures. We reject this null hypothesis and conclude that a diamond of Clarity 4 sells for significantly more than a diamond of Clarity 3.

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## Prediction

Summit Jewelers has a diamond with Carat weight 2.1 and Clarity 6. Summit Jewelers, asks to predict the price for which it can sell the diamond and to form a 95% prediction interval.

```
# Prediction for Summit Jewelers, Carat=2.1 and Clarity = 7
Xnew = pandas.DataFrame({"Carat": [2.1], "Clarity": [6]})
y_pred = LogFacDiam211.prediction_intervals(Xnew)
exp_ypred = y_pred.map(numpy.exp)
exp_ypred
```

We obtain:

| Predicted | Lower | Upper |
|---|---|---|
| 23115.511663 | 19878.82351 | 26879.200329 |

We predict the diamond price to be $23,115 with 95% prediction interval [$19,879, $26,879].

- Note the difference between the upper bound and the predicted value is $3,763.
- The difference between the predicted value and the lower bound is $3,237.

What is the intuition for the difference?

The prediction interval for log price is symmetric about the mean of log price. When we exponentiate, the upper bound increases proportionately more than the lower bound. It is still a 95% prediction interval price.

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## Summary

We studied two diagnostics for functional form.

- The Ramsey test helps us determine whether we need a transformation of our data. It evaluates the function we have chosen. It does not tell us what function to use.
- Leverage plots provide a visual diagnostic that can sometimes help us to determine which transformations we might consider.

Don't feel that you need to have a full grasp of these diagnostics at this point. We'll do applications as we go forward, and you will become increasingly comfortable with using them.

We chose a functional form for the data for Diamonds_211.

We saw how to interpret the coefficient of log(Carat) and we saw how to use the C() command to allow a different coefficient for each value of Clarity.

We saw how to interpret the coefficients of the different values of Clarity.

We did prediction and prediction intervals for a model with dependent variable in log form.

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## Summary of This Set of Slides

This note set introduced study of functional form.

We studied the following functional forms and illustrated with applications:

- Models with one or more variables in log form
- Semi-log form: Dependent variable logged and independent variable not logged
- Quadratic functional form
- Testing functional form using Ramsey test

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## Appendix: More Leverage Plot Detail, 1 of 2

Suppose you have a dependent variable y and multiple right-hand-side variables x1, x2, etc.

- A separate scatter plot of y against each x variable may not be very informative because multiple varies all affect the value of y.

Leverage plots are designed to try to overcome this problem and present information similar to a scatter diagram:

- A leverage plot of y against x1 is designed to show the relationship of y with x1 while controlling for x2, x3, etc.
- Similarly, a leverage plot of y against x2 is designed to show the relationship of y with x2 while controlling for x1, x3, etc.

In short, the leverage plot shows the effect of each right-hand-side variable on y after removing the effects of all other x variables.

A clever strategy lies behind construction of leverage plots as explained on the next slide.

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## Appendix: More Leverage Plot Detail, 2 of 2

Back

We estimate a regression of y against all variables except $X_1$.

The residuals u from this regression retain the effects of $X_1$ on y; the effects of all other X variables have been removed.

We estimate a regression of $X_1$ against all other X's.

The residuals v from this regression retained the variation in $X_1$ that is distinct from variation of the remaining X variables.

- Hence, the leverage plot isolates the relationship of y and $X_1$ after removing the effects of all other variables.
- It can be shown that a regression of u on v gives a slope precisely the same as the coefficient of $X_1$ in a regression of y against all X's.

If we find the leverage plot is not linear, then transforming $X_1$ can often help improve the fit to the data y.

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## Derivation of Elasticity in Log-Log Model

By definition, the elasticity of y with respect to a variable x is: $\frac{x}{y}\frac{\partial y}{\partial x}$

Log-log relationship: $\log(y)=\beta_0 + \beta_1\log(x)+\ldots + \varepsilon$

Differentiating with respect to x: $\frac{1}{y}\frac{\partial y}{\partial x}=\frac{\beta_1}{x}$

Rearranging: $\frac{x}{y}\frac{\partial y}{\partial x}=\beta_1$

This proves that $\beta_1$ is the elasticity of y with respect to x in a log-log relationship.

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## *Derivation: Semi-Log Relationship for Small Change in x*

Consider the semi-log relationship:

$\log(y)=\beta_0 + \beta_1 x + \varepsilon$

Differentiate with respect to x: $\frac{1}{y}\frac{\partial y}{\partial x} = \beta_1$

We can rewrite this putting (dy/y) in the numerator and dx in the denominator: $(\frac{dy}{y}) / dx = \beta_1$

This proves that $\beta_1$ is fraction change in y with respect to a small change in x in a semi-log relationship.

This derivation applies for $\beta_1 \Delta x < .15$

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